

Proof Architecture in the OpenAI Lean Library: Representation, Locality, and Semantic Transport

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Abstract

What information must a change of representation preserve for a formal proof to continue? This review studies that question in six developments from the OpenAI Lean collection: common-base counting, graph switch chains, the symmetric Mahler inequality, the complete Crouzeix bound, extended trace cones, and marked character-variety constructions. We follow the declarations that transport probability laws, computational bounds, analytic estimates, and coordinate identities to their downstream uses. The cases expose a division of proof work: mathematical interfaces establish the hypotheses under which local arithmetic, algebraic, and finite checks suffice. Elementary examples explain the information these interfaces must retain. An exploratory survey compares 800 uniformly sampled OpenAI modules with 8,070 mathematical modules in their pinned Mathlib dependency. With equal weight per proof-bearing module and joint standardization by broad subject and proof-length band, arithmetic and algebra tactics occur in a larger share of extracted named proof bodies by 22.8 percentage points; the differences are 10.4 points for tactic-mode proofs and 5.8 for local claims or calculation chains. Comparing these estimates with subject-only controls separates persistent arithmetic emphasis from differences associated with proof length. Large numerical certificate batches also make pooled body frequencies differ sharply from module averages. Together, source tracing and corpus analysis identify how established Lean methods are organized in this collection and give readers a practical way to inspect the mathematical obligations behind a formal theorem.

Keywords: formalized mathematics; Lean; proof architecture; proof automation; semantic preservation; mathematical interfaces.

1 Introduction

A formal proof often changes the language in which its objects are represented. A graph becomes a Boolean matrix; an algebraic wedge product becomes a continuous bilinear operator; a randomized estimator becomes a program reading addressed tape bits. The next argument needs more than a new description of the object. It needs a theorem preserving the particular quantity or operation on which that argument depends.

We call such correspondences preservation interfaces. An interface specifies the relation between two representations and exposes the information available to a later construction. To transfer an accuracy guarantee, it must preserve the relevant event probability; equality of output laws supplies a reusable sufficient condition. To differentiate a field, it must provide appropriate local

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agreement. To transport an equation in named coordinates, it must identify the coordinate maps. These requirements explain both the mathematical content and the formal shape of the interfaces studied here.

Six developments in the OpenAI Lean collection provide the comparison [39]. Common-base counting and graph switch chains connect finite representations to probabilistic guarantees. The Mahler and Crouzeix arguments connect algebraic identities to differentiation, integration, and norm estimates. Trace ideals and marked character-variety diagrams reconstruct or transport algebraic data. The first four selected statements are public theorem endpoints; the last two are support results. Keeping these roles explicit allows the comparison to follow the mathematical work performed by each selected construction.

Three questions guide each case: what does the intermediate representation retain, which theorem relates it to the intended object, and how is that theorem used in the next step? The case studies answer these questions through source declarations and worked mathematical examples. A separate exploratory survey compares the written proof methods of 800 uniformly sampled OpenAI modules with the pinned Mathlib dependency. It distinguishes average within-module practice from body frequencies dominated by large certificate batches, then examines broad subject and proof-length composition. The two analyses serve mathematicians reading formal source and formalizers designing reusable interfaces.

Together, they describe a division of proof work. Representation and preservation lemmas make the hypotheses of a local calculation available; simplification, arithmetic tactics, or exact decision procedures then check the resulting obligation. Arithmetic and algebra tactics remain substantially more prevalent after joint subject and length controls. The corresponding differences in tactic blocks and intermediate claims are smaller under these controls. Source examples explain how such devices operate within particular proofs: the construction determines which goals reach a tactic and why the available hypotheses suffice.

Companion reviews assess the proposed mathematical advances against prior results [23] and examine counting and sampling applications [22]. The present review uses common bases, Mahler, and Crouzeix to study a different question: how the formal source connects its mathematical representations and organizes the proof obligations between them.

The comparison builds on established accounts of formalization. Lean combines dependent type theory with an extensible elaborator and a programming language [17]; Mathlib supplies a shared mathematical vocabulary [43]. Avigad et al. [2] trace a linear-algebra example through formal proof, generalization, and connection to an existing Mathlib theorem. Commelin and Topaz [13] explain how usable specifications support mathematical construction and collaboration. Here we compare the mathematical payload of preservation statements across several settings and relate their local uses to a corpus-level method profile.

The odd-order theorem and Kepler conjecture establish precedents for large proof decompositions and verified computation [20, 26]. Lean formalizations of perfectoid spaces and schemes illustrate the importance of representation choices [8, 9]. These precedents frame the novelty question in this review: which established device is used, what project-specific information does its interface preserve, and what later step becomes possible because that information is available?

Section 2 describes selection and evidence. Section 3 begins with the overview in Table 1, then develops the six cases. Section 4 compares source-written proof methods with Mathlib, including joint composition controls and certificate concentration. Section 5 isolates the cases' preservation requirements in elementary examples. Section 6 connects the mathematical and empirical findings and develops guidance for new formalizations.

2 Sources, selection, and reading method

Each case follows a chain consisting of an initial representation, a preservation statement, and its downstream use. Selection was purposive: a candidate had to expose all three parts and a mathematically meaningful change of representation. We sought contrasting preservation requirements across mathematical settings so that the comparison would explain why different interfaces need different hypotheses.

We read backwards from each selected conclusion to its definitions and hypotheses, then forwards through the constructions supplying those hypotheses. This separates the established mathematical device, the specific relation proved in the development, and its use in a later argument. Evidence ledgers retain declaration names, source excerpts, line ranges, and Git blob identifiers. Matching downloaded files to these identifiers makes the source locations reproducible.

Discovery proceeded from endpoint inspection to deeper tracing of candidate interfaces. The irrationality-exponent-of- π and two-point-correlation endpoints were considered but excluded because their mechanisms had not been traced to comparable depth. The six-case selection therefore supports a focused comparison of interfaces. Section 4 addresses prevalence separately through a uniform path sample, with explicit extraction units and feature definitions. This division keeps purposive mathematical analysis distinct from estimates about the wider source population.

The key reading task is to distinguish an interface from its instantiation. When an analytic lemma takes a flux identity or resolvent estimate as input, the analysis follows the endpoint's construction of that input. Likewise, a witness selected by classical choice is traced to its existence theorem. This identifies which hypotheses remain in the final statement and which are supplied within the proof.

All six cases are studied through declaration and dependency analysis. Their transitive OAI import closures are retained, and the evidence ledgers identify the bodies inspected. The worked examples in Section 5 include arithmetic and finite-law lemmas checked with Lean 4.34.1. A separate experiment compiles three unchanged upstream trace-cone modules and an importing corollary using the pinned Mathlib revision. The data and code availability statement records the source provenance and the scope of these complementary checks.

Blueprints link informal arguments to formal declarations and dependencies [32]; LeanArchitect studies automation of that connection [47]. Our ledgers serve as source locators for the selected interfaces. The review adds the mathematical explanation: the equations, normalizations, and hypotheses that make each transition useful to the next argument.

3 Six mathematical interfaces

Table 1 groups the cases by probability and computation, analysis, and reconstruction of algebraic data. Each account identifies the representation, the information its bridge preserves, and the next operation using that information. Section 5 isolates the requirements in elementary examples.

Table 1. Overview of the selected formal statements and the information preserved by each bridge.

Development	Representation bridge	Mathematical work exposed
Common bases Endpoint	Matroid transversals; finite random trees; oracle expressions; physical random tapes.	Output-law preservation joins bit-cost bounds and a fixed-program specification.
Graph switches Endpoint	Fixed-degree graph cells; 2×2 Boolean matrices; conditional averaging operators.	A finite count isolates one exceptional direction per fiber; compensated estimates control its global contribution.
Mahler Endpoint	Ordered shuffle forms; continuously bundled fields; boundary-local extensions.	Exact slot identifications preserve derivatives, signs, and multiplicities; locality makes Stokes applicable.
Crouzeix Endpoint	Amplified polynomial evaluation; rectangular compressions; Fourier coefficient energies.	Finite dual witnesses and positive blocks give a scalar cancellation; outer domains remove strict-containment assumptions.
Trace ideals Support result	Closed ideals; idempotent extended traces; positive cuts and scalar limits.	Algebraic recovery of ideals combines with canonical convergence to identify transported ideals.
All seams Support result	A shared splitting witness; named graded maps; rebased matrices and produced diagram values.	Eight local seam families cover the constructed diagram; equality transport retains its named maps.

3.1 Common bases: from a matroid representation to physical random tapes

The common-bases development follows an approximate count all the way to a fixed oracle program, its bit costs, and the tape positions supplying its randomness [35]. Three descriptions of the computation are kept explicit: the mathematical estimator, a finite random experiment, and the compiled machine. Successive interfaces track the counted objects, the output law, and execution on a given input through this pipeline.

Encoding the counting problem. The mathematical problem is to count the common bases of two equal-rank matroids M_1, M_2 on an enumerated ground set $E = \text{Fin } n$. The first representation replaces two simultaneous base conditions by one base condition. On $E \sqcup E$, set

$$N = M_1 \oplus M_2^*, \quad T(A) = \text{inl}(A) \cup \text{inr}(E \setminus A).$$

The source proves that $T(A)$ is a base of N exactly when A is a base of both input matroids.¹ Equal ranks give $r(N) = n$, while every transversal $T(A)$ has size n . Consequently the deficiency $d(A) = n - r_N(T(A))$ vanishes precisely at the objects being counted. Explicit equivalences between subsets, finite subsets, and transversals transfer both cardinalities and partition sums. If Z is the common-base count, the resulting annealing partition function satisfies

$$Z \leq \sum_{A \subseteq E} q^{d(A)} \leq Z + 2^n q, \quad 0 \leq q \leq 1.$$

The partition function now has a direct relation to the desired count: zero-deficiency terms contribute exactly Z , and every other term is at most q . The doubled ground set also supplies the exchange space. A plain state selects one element from each pair; a defect state has one empty pair, one full pair, and singleton choices elsewhere. The walk passes through paired defects before a hitting construction returns it to plain states. This separates the space convenient for local moves from the transversals whose weighted count is estimated.

A greedy routine next computes rank from independence queries. Its correctness theorem identifies the implemented deficiency and annealing weights with the mathematical ones. A sampled zero-deficiency witness serves as the estimator's final guard. Because such a witness is a common base, this guard enforces the zero-count conclusion on every execution, including tapes outside the high-probability accuracy event.

From experiments to a bounded program. Finite random experiments are trees with uniform choices at their internal nodes. The bounded implementation gives each draw a bank of rejection trials. Its fit condition bounds the number of successive draws by M and requires $H \geq \lceil \log_2 a \rceil$ trial bits for each positive draw arity a . With D trials per draw, the observable loss is at most $M2^{-D}$ for values in $[0, 1]$. The width condition ensures that a trial can represent every outcome; increasing the number of undersized trials would not repair that defect. A separate reindexing relation permits permutations of outcome labels and proves expectation preservation. Consequently, the walk and its implementation may enumerate choices differently while agreeing in law.

In the expression language, randomness becomes explicit input. Evaluation takes a value and a random bank and returns an optional result together with the unused bank. A realization theorem relates this deterministic execution, bank by bank, to the random tree. For a Metropolis step, the proof follows the proposal, acceptance calculation, Bernoulli choice, and lazy branch. Composition with the witness guard then gives the complete estimator. Law preservation and bankwise execution preservation enter at different layers of the same construction.

Compilation preserves the encoded output and supplies a cost bound. The constructed nine-register stack machine implements the expression language by structural simulation, including its list folds. Source evaluation returns both a value and a charge; the machine produces that value within the recorded charge. This makes resource use part of the simulation statement, ready for the polynomial-bound argument.²

From semantic preservation to a polynomial cost bound. Bounding the charge requires control of intermediate representations. The recursive predicate $\text{Safe}(O, S, e, x)$, with oracle O , width bound S , expression e , and input x , bounds the encoded widths of the input and every successful output, and requires safety of the subevaluations actually used. For a composition it

¹Base correspondence and the rank of the paired matroid.

²Cost-bearing compiler correctness.

bounds the intermediate result; for a fold it requires the invariant after each accumulator update. If the body charge is at most E and these widths are at most S , the fold lemma gives

$$c_{\text{fold}} \leq |\text{xs}| (7 + 12S + E).$$

The encoding bounds list length by width, so this estimate can be composed through structural induction on the fixed expression. The width certificate turns the cost-bearing simulation into a polynomial charge bound.³

The source packages these certificates as **PolySafe**: a polynomial p takes an input parameter S to an admissible width $p(S)$. Polynomial substitution supports composition. A further interface, **AttrSafe**, carries numerical predicates alongside the width bound. This matters for value-indexed operations: an exponent can be very large despite a short binary encoding. The exponentiation lemma consequently assumes that the exponent itself is at most S . The estimator's attribute bounds three parameter fields, and its proof transports those bounds through exponentiation, range construction, and ratio products. In `attrSafe_compose`, the next width parameter is enlarged to $g(S) = p(S) + S$; monotonicity of the numerical attribute carries its information to that larger parameter. Thus a usable complexity interface preserves both encoded size and the value constraints needed by the next operation.⁴

From output laws to successful tapes. The final probabilistic interface connects the program to its accuracy guarantee. Let V be the canonical bounded experiment, and let v be the rational output function implemented by the postamble. The certificate asserts

$$\mathbb{E}_w \varphi(v(w)) = \mathbb{E}_b \varphi(V(b)) \quad \text{for every } \varphi : \mathbb{Q} \rightarrow \mathbb{R},$$

where both expectations are finite uniform averages. It also records source execution and nonnegativity, including zero output when $Z = 0$.⁵ The universal observable interface transports the output distribution through changes of enumeration and bank layout.

The machine correspondence reaches actual tape addresses. For m total gathered bank bits, there is a start offset s , fixed before quantifying over tapes, such that every tape $\beta : \mathbb{N} \rightarrow \{0, 1\}$ produces

$$v(\text{physicalWord}(s, m, \beta)), \quad \text{physicalWord}(s, m, \beta)(i) = \beta(s + 3(m - 1 - i) + 1), \quad i < m.$$

The reversal comes from pushing bits onto a stack; the stride three reflects the sampling loop's instructions. The proof supplies $s + 3m + 1 \leq B$. Injectivity of these positions shows that a uniform length- B tape induces a uniform length- m word.⁶ Here m counts bank bits, whereas B bounds complete machine execution.

The endpoint quantifies over one fixed program and constants C, a before the input matroids, oracle, and rational parameters $0 < \varepsilon, \delta < 1$. For an exact independence oracle and matroids with full ground set and supplied common rank r , the resource bound is

$$B = C(1 + n + \ell + \lceil \varepsilon^{-1} \rceil + \text{clog}_2 \lceil \delta^{-1} \rceil)^a,$$

where ℓ is the encoded input length and clog_2 is the ceiling binary logarithm. Execution on every infinite tape produces an output within B , with at most B oracle calls and bit operations. For

³See **Safe** and the fold-charge proof, and the structural polynomial bound.

⁴See **PolySafe**, attribute composition, and the exponent and estimator attributes.

⁵The output certificate.

⁶Physical tape positions and their uniform law; the machine correspondence.

a length- B word, let $\text{out}(w)$ denote this output on the word's zero extension. Substituting the accuracy indicator in the observable identity yields

$$\#\{w \in \{0,1\}^B : (1 - \varepsilon)Z \leq \text{out}(w) \leq (1 + \varepsilon)Z\} \geq (1 - \delta)2^B.$$

The specification packages this success count with the universal resource and zero-output guarantees.⁷ Verified compilation and probabilistic semantic preservation are established methods [6, 7, 29, 30]. Expectation-functional semantics for randomized programs is also established [1]. In Lean, SampCert separates sampling procedures from normalized distribution semantics and proves exact sampler laws [16]. Its **SLang** represents a program by an unnormalized mass function; normalization and equalities between sampler laws support downstream privacy reasoning. The common-base development instead carries its bounded random experiment through a costed expression language to a fixed stack machine. A close compiler precedent is Zar, which translates probabilistic programs through choice-fix trees and interaction trees to random-bit samplers, proving equidistribution in Coq [4]. Here the comparison turns on the specification carried through the pipeline: intermediate widths and numerical attributes, an every-tape resource bound, and the finite count of accurate tapes. The distinctive evidence is the concrete integration: the proof carries the counting quantity through oracle code, width bounds, and physical tape coordinates to an exact finite success count. Sections 5.1–5.3 isolate the uniformity, law, and error-budget arguments behind this last step.

3.2 Graph switches: a local table and a fiberwise error direction

The graph case studies how two conditional averages interact. Its switch-chain development connects a four-vertex representation to a spectral estimate [40]. The chain stays put with probability $1/2$; otherwise it chooses uniformly a four-vertex set and an ordered pair of distinct perfect matchings on it. It replaces the first matching by the second when the removed edges are present and the added edges are absent, and stays put otherwise. Each of the $6\binom{n}{4}$ proposals has probability $1/(12\binom{n}{4})$. For $n \geq 4$ and a graphical degree sequence satisfying $d(v) \leq n - 1$, the endpoint gives mixing time at most $2n^8$ at total-variation threshold $1/4$. When there is more than one realization, its spectral-gap bound is

$$\frac{1}{24n^2\binom{n}{4}}.$$

We follow the disjoint-pair part of the argument, where an exact matrix count identifies the exceptional direction that the operator estimate must control.

Reducing a graph cell to a matrix count. States are simple graphs on $\text{Fin } n$ realizing d . Fix four distinct vertices, grouped into pairs $a = \{i, j\}$ and $b = \{k, l\}$. A coarse cell fixes the outer graph, the internal edge of each pair, and every outside vertex's total incidence into each pair. The remaining data consist of a 2×2 Boolean matrix M of cross edges and two sets U, W recording the singleton boundary choices at i and k . The source proves that these coordinates determine the graph.⁸

Once the cell data are fixed, the four degree equations reduce to the cross-edge total and two margin equations:

$$\text{row}_0(M) + |U| = c_L, \quad \text{col}_0(M) + |W| = c_R,$$

where c_L, c_R are determined by the original graph. Forward and inverse constructions identify the graph cell with boundary choices and a compatible matrix. Under this equivalence, a uniform graph

⁷The fixed-program and finite-tape specification.

⁸The coarse graph cell; degree feasibility in coordinates.

law weights each pair of boundary choices by its number of compatible matrices. Preserving these multiplicities is what makes the coordinate model usable for conditional averaging. The matrix count describes the remaining dependence between the left and right choices.

There are only sixteen Boolean matrices. Let $c_s(r, t)$ count those with total s , first-row sum r , and first-column sum t . For $s = 2$, finite decision proves the table

$$(c_2(r, t))_{r, t=0}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad c_2(r, t) = 2 - p(r) - p(t),$$

where $p(r) = \mathbf{1}_{r \neq 1}$.⁹ The other totals give tables that factor into independent margin weights. For total two, centering leaves dependence through one Boolean flag on each side. This reduces the correlated component to a specified local direction, which the subsequent operator argument can isolate.

A short calculation illustrates the exceptional correlation. Among the six uniformly chosen total-two matrices, each off-center margin flag has mean $1/3$, while their product is always zero. Their covariance is therefore $-1/9$. The sign shows why a direct nonnegative-correlation argument is unavailable. Actual graph cells also carry boundary multiplicities, so the proof transports the flag description to those weighted fibers before estimating its contribution.

Paying for the exceptional correlation. In the finite coordinate model, composing the two conditional averages gives $A + Bp(r)$ on each fixed cell. The coefficients A, B depend on that cell and the averaged function, and $B = 0$ unless $s = 2$. The graph-cell equivalence transports this formula back to actual graph states. This transport includes a proof that corresponding fibers have equal cardinalities and corresponding sums, and an identification of the flag: $p(r)$ says that i has equal adjacency bits to k and l , meaning both cross edges are present or both are absent.¹⁰

Work in the real function space on the finite graph state space Ω , with counting inner product $\langle f, g \rangle = \sum_{G \in \Omega} f(G)g(G)$. Fiber averages themselves are divided by the fiber cardinality.¹¹ Write E_a for averaging over a pair-resampling fiber. The transported formula expresses the relevant conditional

weight is supported where the cross total is two. For this purpose a fiber is active when k and l each have exactly one neighbor in a . Subtracting E_a leaves a multiple of the centered flag on each active fiber. These functions form an error subspace whose orthogonal projection is denoted R_{ab} . The description is at most one dimensional on each fiber; the complete graph-space subspace can have many independent fiber components.

The flag representation now supplies the hypotheses of an abstract projection inequality. Conditional averaging is symmetric and idempotent, and the error projection fixes the correlated fluctuation. Decomposition into the projections' ranges and orthogonal complements turns these facts into an interaction bound. If G_{ab} denotes the switch-gain projection, the graph instance gives

$$2\|G_{ab}f\|^2 - \|R_{ab}f\|^2 - \|R_{ba}f\|^2 \leq 2\langle (I - E_a)f, (I - E_b)f \rangle.$$

To sum this inequality, the proof needs a budget for the error terms. Fix a graph G and a pair $a = \{i, j\}$. Its singleton-neighbor set V_G consists of outside vertices adjacent to exactly one endpoint of a . The vertices assigned to i form a subset of prescribed cardinality q . An equivalence therefore

⁹The finite total-two count.

¹⁰The affine conditional image; its graph interpretation.

¹¹See `OrthogonalPairResampling.lean`, lines 191–193, and lines 231–232.

identifies this assignment space with the slice $\{S \subseteq V_G : |S| = q\}$. Under its induced linear isometry, the centered equality flag becomes the slice's **raw** vector, while a graph switch becomes a transposition of two labels.

The slice estimate is itself an operator comparison. Normalized flag vectors have squared norm $1/2$ whenever nonzero. Their synthesis operator T satisfies $TT^* \leq L_{\text{swap}}$ in quadratic-form order. The proof decomposes coefficient data into a zero-sum linear part and a symmetric part with zero rows and diagonal. The Gram operator is scalar on these parts; its scalars are bounded by the corresponding swap-generator scalars. Boundary slices have zero flag vectors and are treated separately. The decomposition controls the overlap between flag directions. In projection form it gives

$$\frac{1}{2} \sum_{k,l} \|\Pi_{kl} h\|^2 \leq \frac{1}{8} \sum_{k,l} \|h - h \circ \tau_{kl}\|^2,$$

where Π_{kl} projects onto the centered flag line and τ_{kl} swaps the two labels.¹²

Two normalizations connect this estimate to the graph budget. Multiplying it by two gives a coefficient $1/4$ on the squared differences, exactly the squared norm of the switch-gain projection $(I - S)/2$ for the switch involution S . Next, the restrictions h_G to assignment spaces satisfy the exact identity

$$\sum_{G \in \Omega} \frac{\|h_G\|^2}{\binom{|V_G|}{q}} = \|h\|^2.$$

Each fiber is represented once by each of its graph states; division by its cardinality cancels that repetition. Applying this identity to both sides of the assignment budget lifts the local inequality to the whole graph space. Finally, ordered partner labels (k, l) count each unordered pair twice, so both error and gain sums acquire the same factor $1/2$. The resulting budget bounds the sum of squared error norms by the sum of squared gain norms.¹³ Summing the compensated inequality over ordered disjoint two-element vertex sets a, b then gives

$$\sum_{a,b: a \cap b = \emptyset} \langle (I - E_a)f, (I - E_b)f \rangle \geq 0.$$

The compensated local bounds therefore give a nonnegative aggregate interaction even though individual fibers can exhibit negative correlation.¹⁴

Triple-generator and local comparison estimates combine with the disjoint sum to give a generator-square bound, from which the stated gap and mixing estimate follow. Finite computation and transport through equivalent representations are established proof methods [12, 27]. The informative integration here is the route from a sixteen-object count to a global estimate: identify the equality flag, transport its centered direction with the correct multiplicities, and pay for its aggregate energy through switch gains.

3.3 Mahler: ordered forms and boundary locality

The symmetric Mahler development connects a local differential-form calculation to a convex-body inequality. Exterior algebra describes the calculation, continuous alternating maps support its

¹²See the two-block operator comparison, and isometric reindexing and the swap-sum estimate.

¹³The restriction identity and its use are in `PairEqualityBudget.lean`, lines 151–270; the ordered/unordered conversion is in `PairEqualityBudgetUnordered.lean`, lines 23–54.

¹⁴The concrete interaction inequality; assembly with the projection budget.

differentiation, and boundary integrals supply the quantity used in the geometric reduction. The selected endpoint, `OAI.SymmetricMahler.symmetric_mahler`, states

$$\frac{4^n}{n!} \leq |K| |K^\circ|$$

for $n \geq 1$ and a compact, convex, origin-symmetric set $K \subseteq \mathbb{R}^n$ with nonempty interior. Here volume is ordinary Lebesgue volume and the coordinate polar is literally $K^\circ = \{p : \sum_i p_i v_i \leq 1 \text{ for every } v \in K\}$. The accompanying argument uses holomorphic maps associated with finite intersections of strips [38]. We follow the transgression identity through its regularity and locality requirements to the flux value consumed by the convex-body reduction.

The analytic inputs and the transgression identity. The generic analytic results take a structure named `MassHypotheses`. Its fields require an open domain containing zero, holomorphic components f_j , zero as their unique common zero in the domain, a homogeneous degree- m leading polynomial map G without nonzero common zeros, an actual Euclidean Taylor remainder $f - G = O(\|z\|^{m+1})$, and compact closures of sublevels with $0 < R < 1$ contained in the domain. These fields record analytic inputs from which the mass and flux conclusions are derived. For the strip maps, `euclideanSpecial_massHypotheses` constructs every field from positive dimension, an injective real measurement matrix, and $m \geq 2$. The body theorem passes the resulting homogeneous flux value to the reduction lemma, leaving exactly the stated body hypotheses.¹⁵

For the transgression calculation in dimension $N \geq 2$, write $N = k + 2$ with $k \geq 0$, let $q(z) = z$, and set $\tau(H)(z) = \sum_j |H_j(z)|^2$. The source convention is $d^c u(z)[v] = -Du(z)[iv]/4$. On punctured space the degree- m homogeneous map G defines

$$\beta = d^c \log \tau(G) - m d^c \log \tau(q), \quad \alpha_t = m d^c \log \tau(q) + t\beta.$$

The deformation uses the ordered primitive

$$\eta_t = \alpha_t \wedge \beta \wedge (d\alpha_t)^k.$$

The ambient differentiation theorem preserves both terms:

$$d\eta_t = \beta \wedge (d\alpha_t)^{k+1} - \alpha_t \wedge d\beta \wedge (d\alpha_t)^k.$$

It follows from actual Fréchet differentiation, exterior product rules, and $d^2\alpha_t = 0$. Restriction to sphere tangents then kills the first term. On sphere tangents, the remaining variation of $\alpha_t \wedge (d\alpha_t)^{k+1}$ contains $k + 1$ placements of $d\beta$. Denote this remaining sum by residual_t . Its multiplicity gives the identity

$$\text{residual}_t = -(k + 1)d\eta_t \quad \text{on sphere tangent tuples.}$$

The two identities distinguish ambient differentiation from its restriction to the integration boundary. The sign and the factor $k + 1$ identify the residual as an exact form there.

The disappearance of the first term has a concrete linear-algebra proof. At $z \neq 0$, the sphere tangent space has real dimension $2N - 1$ and contains the nonzero circle tangent iz . Homogeneity gives $\beta_z(iz) = 0$ and $(\alpha_t)_z(iz) = m/2$. Rotation invariance, together with this constant circle evaluation, gives $d\alpha_t(iz, v) = 0$. Thus β and $d\alpha_t$ are horizontal in this direction: inserting iz in any

¹⁵The endpoint and concrete hypothesis construction are `Mahler/MainTheorem.lean`, lines 12–22, and `Mahler/EuclideanSpecialMap.lean`, lines 67–97.

slot gives zero.¹⁶ Their wedge has top degree on the sphere tangent space and therefore vanishes. The lemma `horizontal_top_eq_zero` proves the general assertion explicitly. An independent top-degree tuple determines a basis. Choose a nonzero basis coordinate of the horizontal vector, replace the corresponding tuple entry by that vector, and expand it in the basis. Alternation kills every term except the selected coordinate times the original evaluation; cancelling the nonzero coordinate proves that evaluation zero. A dependent tuple already vanishes by alternation. The sphere theorem instantiates this argument with the verified tangent dimension and iz .

Identifying two representations of the form. The development constructs η_t in two ways suited to different operations. A literal shuffle form is bundled through finite basis expansion, supporting regularity proofs by evaluation. A continuous product of a two-form and wedge powers supports differentiation rules. Recursive slot types retain each factor’s ordered pair of arguments. Two equivalences flatten these slots to $\text{Fin}(2(k + 1))$ along different routes. A small extensional proof identifies the equivalences; injectivity of the projection from continuous alternating maps to alternating maps then identifies the entire fields. Equality of the whole fields transports the transgression into the analytic theorem expecting the other representation. The intermediate theorem `homogeneousFlux_endpoints_of_transgression` has an explicit local-formula premise, which `homogeneous_flux_endpoints` discharges.¹⁷

The shuffle description exposes permutation signs and factor order for algebraic rewrites. The continuously bundled field has a normed-space structure, so the calculus library interface can express its regularity and derivative. Finite basis expansion connects these obligations: smoothness of the finitely many evaluated coefficients supplies smoothness of the bundled form. The source obtains every finite regularity order on punctured space from the analytic hypotheses, including the C^2 order used below. Identifying the fields also identifies their derivatives as functions valued in continuous alternating maps. Agreement on tangent tuples supplies the later integration comparison; differentiation uses equality of the form-valued fields. The proof therefore uses two distinct congruence principles at appropriate stages: whole-field equality for differentiation and tangential agreement for oriented density. The source uses each principle at the stage where its information suffices.

The continuous product also exposes a useful implementation choice. `continuousWedge` starts with the algebraic shuffle product and proves continuity in its vector arguments term by term. For continuity as a function of the two forms, the source first embeds continuous alternating maps into multilinear maps. Injectivity gives finite dimensionality of the form space when the ambient real space is finite dimensional. It then bundles the product as

$$\mathcal{W} : \mathcal{A}_\iota \longrightarrow_L (\mathcal{A}_\kappa \longrightarrow_L \mathcal{A}_{\iota \sqcup \kappa}),$$

where $\mathcal{A}_\iota$ is the normed space of continuous alternating maps indexed by ι , and $\longrightarrow_L$ denotes a continuous linear map over $\mathbb{R}$. The two applications of `LinearMap.toContinuousLinearMap` use finite dimensionality to promote algebraic linearity to continuity. The regularity of $x \mapsto \mathcal{W}(a(x))(b(x))$ now follows from the ordinary calculus of continuous linear maps and their application.¹⁸ The finite-dimensional setting makes this bridge economical: the bundled bilinear product supplies the calculus operations, and the shuffle representation supplies exact algebraic identities. Mathlib already supplies the exterior derivative and its square-zero theorem [44]; the project-specific work here connects the ordered product, its regularity, and the integration interfaces.

¹⁶The normalization and circle evaluations are in the d^c definition, lines 12–13, and the deformation and circle identities, lines 14–101.

¹⁷See the homogeneous endpoint, lines 12–45. The underlying tangent identity is proved in the tangent transgression theorem, lines 65–97.

¹⁸See `ContinuousWedge.lean`, lines 19–107.

From complex forms to real-coordinate flux. The integration interface requires another preservation theorem. The primitive is a complex-valued real alternating form, whereas coordinate Stokes takes real-valued forms on $\mathbb{R}^{2N}$. Let $L : \mathbb{R}^{2N} \rightarrow \mathbb{C}^N$ be the positively oriented orthonormal coordinate map used by the sphere flux. The source defines the real pullback

$$(\mathcal{P}\eta)_x(v_1, \dots, v_q) = \operatorname{Re}(\eta_{Lx}(Lv_1, \dots, Lv_q))$$

and proves $d(\mathcal{P}\eta) = \mathcal{P}(d\eta)$ wherever η is differentiable. Taking real parts commutes with the finite alternating derivative sum; the chain rule handles the fixed linear coordinates. A separate density identity uses the chosen frame’s volume $+1$, and a measure-preserving sphere isometry identifies the two flux integrals. Thus the exact-form identity reaches coordinate Stokes with its sign and normalization preserved.¹⁹

Applying Stokes near the boundary. A second interface controls locality. The primitive is C^2 on punctured space, while the flux argument needs regularity near the sphere. `exists_close_d_set_form_extension` takes a form ω that is C^2 on an open set U containing a closed set K . Smooth separation gives a function χ equal to one near K and zero near U^c . The field $\tilde{\omega} = \chi\omega$ is globally C^2 : use the product rule inside U and its eventual equality to zero outside U . Near K it is eventually equal to ω , so their Fréchet derivatives, and hence exterior derivatives, agree there. For K the sphere, coordinate Stokes applies to $d\tilde{\omega}$ and $d^2\tilde{\omega} = 0$ gives zero boundary flux. Tangent-density congruence then transfers that conclusion to the original residual.²⁰

Zero residual flux makes the deformation’s path flux constant by the fundamental theorem of calculus. Comparing endpoints supplies $(\pi m)^N$; the small-sphere limit and special strip-map reduction use that value in the body inequality. Dimension one is handled by a separate circle calculation. The mathematical devices are familiar. Mathlib already provides continuous alternating maps, exterior differentiation, and $d^2 = 0$ under smoothness assumptions [44]. Kudryashov’s earlier Lean formalization already transfers a divergence theorem between integration theories and derives Cauchy–Goursat under carefully chosen differentiability assumptions [28]. Its rectangular domains give a direct comparison for integration interfaces. The integration here combines equality of ordered and continuously bundled forms with a locality theorem supplying regularity near the boundary. This connects the analytic calculation to Stokes while accommodating the singular center. Section 5.4 illustrates why the ambient and tangential comparisons need different data.

3.4 Crouzeix: compression, energy, and strict neighborhoods

The Crouzeix case follows a different analytic interface: boundary coefficient estimates feed a matrix-norm calculation with constant two. For an n -by- n complex matrix A , m -by- m coefficients B_k , and $n, m > 0$, `OAI.DirectCrouzeix.complete_crouzeix` states

$$\left\| \sum_{k=0}^d A^k \otimes B_k \right\| \leq 2 \sup_{z \in W(A)} \left\| \sum_{k=0}^d z^k B_k \right\|.$$

Write $F(z) = \sum_{k=0}^d z^k B_k$; $F(A)$ denotes the amplified evaluation on the left. The source uses the operator 2-norm, literal matrix Kronecker products, and $W(A) = \{\langle u, Au \rangle : \|u\| = 1\}$. Here C^* denotes the conjugate transpose, and the Hilbert–Schmidt (Frobenius) norm is $\|C\|_{\text{HS}}^2 = \sum_{i,j} |C_{ij}|^2$.

¹⁹See real-part projection and pullback naturality, lines 33–96, and the oriented density and flux comparisons, lines 11–67.

²⁰The extension and exact-form flux theorem are `Mahler/ClosedBoundaryStokes.lean`, lines 18–63.

Its endpoint quantifies over arbitrary A and coefficient matrices. The accompanying argument [37] is organized around boundary analysis. The mathematical-frontiers review explains its ordered product tests and positive-block closure in relation to earlier proof routes [23]. Here the focus is the formal implementation: zero-mode normalization, the finite dual witness, coefficient transport, and construction of the strict neighborhoods used by the endpoint.

Closing the compressed energy estimate. The local interface `analytic_domain_core` accepts a polynomial sequence b_k with $b_0 = 1$, a continuous boundary parametrization, and a matrix resolvent field $R(t)$. Its hypotheses give finite Faber expansions, lower and upper coefficient-energy bounds, exact coefficients $\widehat{R}(-k) = b_k(A)$, vanishing positive Fourier coefficients, and $R + R^* \geq 0$. These are the inputs of the local theorem. `exterior_collar_bound` constructs them from exterior conformal data and supporting halfplanes before applying the core. In particular, the positive resolvent and coefficient bounds are constructed at this geometric stage before the public endpoint is assembled.

For the normalized core, assume $\|F(h(t))\| \leq 1$ on the boundary, where h is its parametrization. Let T be the amplified polynomial evaluation and $\gamma = \|T\| > 0$; the zero case is separate. A unit singular pair satisfies $Tx = \gamma y$ and $T^*y = \gamma x$. The supporting lemma obtains a norm attainer by compactness, then proves the adjoint relation through a squared-distance calculation. Reshaping vectors indexed by $\text{Fin}(n) \times \text{Fin}(m)$ produces rectangular matrices X, Y . All boundary integrals below use normalized Haar measure μ on the period-one circle $\mathbb{R}/\mathbb{Z}$. The proof forms

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY, \quad a = \int \|M(t)\|_{\text{HS}}^2 d\mu(t).$$

A cross-boundary pairing estimate makes $a = 0$ incompatible with $\gamma > 0$. The two singular relations then produce the diagonal bounds

$$\int \|P + P^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}, \quad \int \|Q + Q^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}.$$

The normalization $b_0 = 1$ identifies the zero Fourier coefficients:

$$\widehat{P}(0) = X^*X, \quad \widehat{Q}(0) = Y^*Y, \quad \widehat{L}(0)^* = \widehat{M}(0).$$

The first two are Hermitian; the last is the zero-frequency premise of the cross-energy comparison. All four compressions inherit the vanishing of the positive Fourier coefficients of R . Fourier identities then give $a \leq \int \|M + L^*\|_{\text{HS}}^2$. Compressing $R + R^*$ by the concatenated columns $[X \ Y]$ yields a positive block matrix with these diagonal and off-diagonal blocks. For a positive block matrix $D = \begin{pmatrix} U & Z \\ Z^* & V \end{pmatrix}$ and $J = \text{diag}(I, -I)$, positivity of D and JDJ gives

$$0 \leq \text{Re tr}(DJDJ) = \|U\|_{\text{HS}}^2 + \|V\|_{\text{HS}}^2 - 2\|Z\|_{\text{HS}}^2.$$

Integrating this estimate gives

$$2 \int \|M + L^*\|_{\text{HS}}^2 \leq \int \|P + P^*\|_{\text{HS}}^2 + \int \|Q + Q^*\|_{\text{HS}}^2.$$

Combining the inequalities gives $2a \leq 8a/\gamma^2$. Since $a > 0$ and $\gamma > 0$, multiplication by γ^2 and cancellation yield $\gamma^2 \leq 4$, hence $\gamma \leq 2$.²¹

²¹The four compressions and closing cancellation appear in `DirectCrouzeix/DomainCore.lean`, lines 15–117.

Where the sharp energy weights come from. Two ingredients fix the constants in this reduction. First, `singular_diagonal_estimates` tests both ordered products FG and GF , where G is a test polynomial. The forward and adjoint singular relations turn them into different diagonal compressions, preserving the noncommutative order of matrix coefficients. The dual step is explicit: from bounds on every finite test family C_k , `weighted_coefficient_dual` substitutes $C_k = w_k^{-1} \overline{P_k}$ in the project's bilinear entry pairing, where the bar denotes entrywise complex conjugation. Both pairing and test energy become $s = \sum_k \|P_k\|_{\text{HS}}^2 / w_k$, leaving $s^2 \leq bs$ and hence $s \leq b$. Nonnegative bounded partial sums subsequently give the infinite sequence estimate. The infinite bound is thus obtained from explicit finite optimizing witnesses.

Second, the weights are derived by boundary analysis. A Faber series splits into a nonnegative Fourier series and a remainder with no nonnegative coefficients. Orthogonality gives its energy as the sum of coefficient energy and remainder energy. The remainder is also the action of a nonnegative kernel with both marginals one on the nonconstant series. A weighted squared-norm inequality and Fubini prove kernel contraction. Consequently $w_0 = 1$ and $w_k = 2$ for $k > 0$. The resulting half-weighted coefficient tail fits the exact Fourier identity

$$\int \|P + P^*\|_{\text{HS}}^2 = 4 \left(\|\hat{P}(0)\|_{\text{HS}}^2 + \frac{1}{2} \sum_{k \geq 1} \|\hat{P}(-k)\|_{\text{HS}}^2 \right).$$

Its factor four at frequency zero uses Hermiticity of $\hat{P}(0)$; the negative coefficients contribute in paired positive and negative frequencies.²²

A further interface supplies matrix-valued Faber expansions from scalar polynomials. A scalar sequence with its k th polynomial of degree k is mapped into polynomials over the matrix coefficient ring by sending scalars to scalar matrices. Its leading coefficients are units. The implementation then applies Mathlib's degree-filtered spanning theorem for polynomial sequences and extracts a finite matrix-coefficient witness from membership in that span. This gives the exact finite expansion consumed by the core. Evaluation and Kronecker-evaluation identities carry the finite witness to the two matrix calculations. The established polynomial basis theorem is reused across the change of coefficient ring, while explicit transport lemmas preserve the order of multiplication. Together with rectangular reshaping, this explains how the scalar boundary analysis supplies estimates uniformly in the matrix coefficient size, the defining feature of the complete bound.

Constructing the analytic inputs. The analytic inputs of the core are built before the final limiting step. For an exterior parametrization $\psi(\zeta) = c\zeta + g(\zeta^{-1})$, the source uses the algebra-valued generating function

$$\mathcal{G}_A(u) = (c - u^2 g'(u))((c + ug(u))I - uA)^{-1}.$$

Its k th Taylor coefficient is the Faber polynomial b_k evaluated at A . The proof makes this coefficient identity finite: a geometric expansion of the inverse has a remainder $u^{k+1}H(u)$ with H analytic near zero, whose k th coefficient vanishes. Local equality transfers the Taylor coefficients. On the analytic disk supplied by the collar construction, evaluation at $u = e^{-2\pi it}$, with $t \in \mathbb{R}/\mathbb{Z}$, invokes the antianalytic Fourier formula, giving the negative modes $b_k(A)$ and zero positive modes. Thus the orientation and normalization consumed by `analytic_domain_core` arise from a proved coefficient calculation.²³

Constructing ψ requires a further change of representation. The source inverts the exterior of a convex exponential sublevel about an interior point, obtains a bounded simply connected domain,

²²See `BoundaryEnergy.lean`, lines 57–88, and `PositiveBlocks.lean`, lines 22–62.

²³See `FaberCalculus.lean`, lines 94–163.

and constructs its Riemann map. To obtain an analytic collar, it first extends that map locally through analytic boundary charts. The gluing argument retains more than analyticity: each local extension E agrees with the original map inside the domain, has nonzero derivative at its boundary point, and satisfies

$$\|E(z)\| < 1 \iff z \in U.$$

This side condition carries the geometric meaning of the unit circle through continuation.

The local-to-global step is explicit in Lean. Extensions are available on balls of radius $3r$, while the chosen cover uses balls of radius r . When two smaller balls overlap, their larger intersection is convex and meets U ; the identity theorem extends their agreement from the interior to that intersection. A classically chosen extension at each point is therefore locally equal to any covering representative. This neighborhood equality supplies analyticity and derivative agreement for the glued map. Compactness and local invertibility then give an injective neighborhood of $\bar{U}$. Its image contains the closed unit disk, so a thickening contains a disk of radius $R > 1$. The inverse on that disk is analytic, and inversion back to the original plane gives the exterior Laurent form.²⁴ The collar therefore retains analytic germs, the domain-side identity, and injectivity. These are the geometric facts needed to justify the resolvent and coefficient estimates used earlier.

Removing the auxiliary domain assumptions. The last step supplies strict-domain geometry for an arbitrary numerical range. For every compact, nonempty convex $K \subseteq \mathbb{C}$, `exists_exponential_outer` constructs a compact sublevel of a finite sum of exponentials of affine support inequalities. A finite direction net controls approximation; four coordinate directions ensure boundedness. The sublevel contains K strictly and lies in any prescribed neighborhood of K . In the endpoint, set $M = \sup_{W(A)} \|F\|$ and $\rho = M + \varepsilon/2$. Compactness places such an outer sublevel inside the open set $\{z : \|F(z)\| < \rho\}$. The scaled domain estimate gives $\|F(A)\| \leq 2M + \varepsilon$, and removing $\varepsilon > 0$ gives the desired bound. The auxiliary outer domain supplies smooth strict containment, leaving positive matrix sizes and arbitrary coefficient matrices in the endpoint.²⁵

The Crouzeix–Palencia proof centers the Cauchy transform of conjugate holomorphic functions and obtains the complete constant $1 + \sqrt{2}$ [14]. Its established bound also supplies the baseline in the companion numerical-range certificate study, which develops finite enclosures, error budgets, and a rational software prototype [24]. The present case follows the identities and geometric hypotheses that connect the formal proof’s local analytic core to its public matrix theorem.

3.5 Recovering ideals from extended traces and canonical limits

The final two cases concern reconstruction and coordinate transport. The trace-cone development recovers closed ideals from extended traces, then identifies the images of two ideals attached to a given trace. The mathematical setting is the cone of extended, lower-semicontinuous traces, studied as an invariant of C^* -algebras by Elliott et al. [18]. The selected support results concern this ideal correspondence. The setting is nonunital complex C^* -algebras with compatible order and star structures, allowing nondensely defined extended traces [41].

Recovering the ideal order algebraically. An extended trace on A , represented by Lean’s `ExtendedTrace` type, is a function on positive elements with values in $[0, \infty]$, together with proofs

²⁴See compatibility and gluing of boundary germs, the inverse collar construction, and the exterior map for exponential sublevels.

²⁵See `DirectCrouzeix/OuterApproximation.lean`, lines 59–122, and `CompleteBound.lean`, lines 65–97.

of vanishing at zero, additivity, positive homogeneity, traciality, and lower semicontinuity. For a closed two-sided star ideal I , the code constructs the extended trace

$$\iota_I(a) = \begin{cases} 0, & a \in I, \\ \infty, & a \notin I, \end{cases} \quad a \geq 0. \quad (1)$$

To establish additivity for positive a, b , heredity of closed ideals gives $a + b \in I$ exactly when both summands belong to I , proving additivity. The equivalence $x^*x \in I \iff x \in I$ and star invariance give traciality; closed sublevel sets give lower semicontinuity. The existence theorem in `FiniteIdeal.lean` supplies all fields of this trace, and the later definition selects the resulting witness.

The reverse construction starts with the finite-square domain of a trace:

$$\mathcal{N}_\tau = \{x \in A : \tau(x^*x) < \infty\}, \quad I_\tau = \overline{\mathcal{N}_\tau}^{\|\cdot\|}. \quad (2)$$

The source proves that $\mathcal{N}_\tau$ is a nonunital star subalgebra absorbing multiplication on both sides, then takes its topological closure. This realizes I_τ as a closed ideal of the original algebra. If $\tau + \tau = \tau$, every value is either zero or infinity. Finite-trace positive elements approximate every positive element of I_τ ; lower semicontinuity extends their zero values to that limit, yielding $\tau = \iota_{I_\tau}$. Conversely, $I_{\iota_I} = I$. These inverse laws establish `closedIdealEquivIdempotent`, an equivalence between closed ideals and additive-idempotent traces.

The ideal order is also encoded by addition:

$$I \subseteq J \iff \iota_I + \iota_J = \iota_I. \quad (3)$$

Consequently, any additive bijection e of the two extended-trace cones already induces an order isomorphism Φ of their closed ideals, with $\iota_{\Phi(I)} = e(\iota_I)$. Additivity and bijectivity suffice for this algebraic reconstruction. Convergence and scalar compatibility enter in the next step, which identifies the ideals attached to a particular trace.

Identifying the transported ideals by limits. The analytic step uses a cut-regularized notion of convergence. Write $(a - \varepsilon)_+$ for the positive cut defined by continuous functional calculus. The development proves

$$a \in I_\tau \iff \forall \varepsilon > 0, \quad \tau((a - \varepsilon)_+) < \infty, \quad a \geq 0. \quad (4)$$

Its canonical convergence predicate for a filtered family (τ_i) and a candidate limit σ requires, for every positive a and every $\varepsilon > 0$,

$$\limsup_i \tau_i((a - \varepsilon)_+) \leq \sigma(a) \leq \liminf_i \tau_i(a). \quad (5)$$

The two inequalities have different roles: the left regularizes the argument by a positive cut; the right uses the original element. Their combination, together with recovery of a lower-semicontinuous trace from its cuts, gives uniqueness of a canonical limit along a proper filter.²⁶

In this sense the source proves

$$r\tau \longrightarrow \iota_{I_\tau} \quad (r \downarrow 0), \quad r\tau \longrightarrow \iota_{I_\tau} \quad (r \rightarrow \infty), \quad (6)$$

²⁶The convergence predicate is defined in `TraceCone/Basic.lean`, lines 65–69; uniqueness is proved in `IdealTransport.lean`, lines 177–190.

where Z_τ is a closed ideal whose positive elements are exactly those on which τ vanishes. The first limit depends on the closure and cut characterization in (4). Whenever $a \in I_\tau$ but $\tau(a) = \infty$, every positive scalar multiple still has value $(r\tau)(a) = \infty$, although $\iota_{I_\tau}(a) = 0$. Finite values on all positive cuts make the scalar-zero limit compatible with (5). The scalar-infinity limit is first proved pointwise using the trace that is zero on τ 's zero set and infinity elsewhere.

The ideal weights, cut-regularized topology, and these two scalar limits are classical parts of the trace-cone analysis of Elliott et al. [18, Sections 3.1–3.2]. The source makes their formal dependencies explicit: additive structure recovers the ideal order, while convergence and scalar compatibility identify the two ideals attached to an arbitrary trace. This separates the algebraic reconstruction from the analytic recognition of the transported objects. Section 5.5 gives a sequence-space example in which the canonical scalar-zero limit and the pointwise limit differ.

Extending convergence preservation to arbitrary nets. The convergence interface makes a further choice about its index type. The source predicate `PreservesCanonicalConvergence`, applied to e , quantifies over filters on the trace cone itself. The mathematical use, however, includes scalar-indexed families $g : J \rightarrow \text{ExtendedTrace}(A)$ and filters l on J . The source passes to the image filter g_*l . Write $\text{CanConv}(l, f, \sigma)$ for the canonical convergence condition in (5), along l for the family f with limit σ . For a further map f into an extended-trace cone, the reindexing lemma states

$$\text{CanConv}(g_*l, f, \sigma) \iff \text{CanConv}(l, f \circ g, \sigma),$$

because the defining $\limsup$ and $\liminf$ commute with this filter map. Applying the preservation interface to the identity family on the trace cone, then reindexing back, proves preservation for g . The lemma `PreservesCanonicalConvergence.net` implements this route. A condition stated using filters on one carrier therefore supports scalar-indexed families and nets on other index types. The filter-map identity is the reusable bridge between these formulations.²⁷

Finally, a `TraceConeEquiv` packages an additive bijection, preservation of zero and positive scaling, and canonical convergence in both directions. For each scalar limit, the endpoint proof maps the family through e , rewrites using scalar compatibility, and invokes uniqueness of canonical limits. Injectivity of $I \mapsto \iota_I$ then yields

$$\Phi(I_\tau) = I_{e(\tau)}, \quad \Phi(Z_\tau) = Z_{e(\tau)}.$$

The displayed endpoint proofs use forward convergence preservation; the inverse field belongs to the larger equivalence interface. The construction thus moves from ideal weights through algebraic reconstruction to analytic recognition by limits. Each stage supplies a particular piece of the identifying equation for the transported ideal. This is a specific use of the abstraction boundaries familiar from large formalizations [8, 13].

3.6 Named graded maps across all seams of a produced diagram

The character-variety case proves that values produced by a marked surface construction satisfy every seam equation. The selected theorem, `producedMarkedValues_allSeams`, starts with an existing field-valued solution of the original diagram. Its hypotheses include properness of the diagram, a maximal-rank condition on the facet at the chosen seam, and positive genus of the parent facet. The conclusion refers to the specific diagram and values constructed from that solution [33]. The key issue is preserving named graded data while changing coordinates and subdividing the surface.

²⁷See filter reindexing and the net extension, lines 238–257.

The diagram consists of surface pieces, called facets, joined along seams; frames specify the graded identifications at those joins. At the selected seam, the construction inserts a refined band determined by a simultaneous splitting of the two input frames. It retains the old boundary circles, appends a mirror boundary at the parent facet, and introduces a genus-zero disk for each fresh strip; the parent genus drops by one. The all-seams theorem checks compatibility of the matrix values on this resulting combinatorial diagram. This roadmap explains the inherited, mirror, and branch families used below.²⁸

The information in a named graded map. The seam equation is stronger than equality of flags. For a graded index set with grade function a and a frame e , let

$$F_k^e = e\{u : u_i = 0 \text{ whenever } a(i) \geq k\}, \quad q_k^e(v) = (e^{-1}v)|_{\{i:a(i)=k\}}.$$

Here q_k^e retains the named coordinates of grade k . The predicate `SameFramedFlag`(a, e, f) requires

$$F_k^e = F_k^f, \quad q_k^e(v) = q_k^f(v) \quad \text{for all } v \in F_{k+1}^e, \quad (7)$$

for every k . Thus the comparison retains the maps representing the successive graded pieces, rather than only their dimensions or underlying subspaces.²⁹ In the produced seam equations, the two frames are obtained from evaluations of the seam's left and right matrix words. Losing these named maps during a change of basis would weaken the theorem's meaning. The one-dimensional frame example in Section 5.6 illustrates this distinction without the surface construction.

Keeping one witness across the construction. The intermediate construction records the shared coordinate data needed by the seam predicate. An `IdentifiedBand` contains a common frame for a simultaneous splitting, a rank shape, and row and column identifications of its graded fibers. Its prefix and named-coordinate laws relate those identifications to the two input frames. The source first proves that such a record exists using the named band grades and rank equalities. It then chooses a witness. The opaque declaration `cutGeneratorWitness` names this chosen construction. Its existence-based selection supplies coherent, noncomputable frame data for the later definitions.

The same chosen record supplies the shape, row ranks, column ranks, and marked frames of the produced diagram. Those fields also determine its punctures and values. Independently choosing compatible dimensions would not provide these shared identifications. The specification therefore links the dimensions to the actual coordinate identifications throughout the construction. Records and algebraic interfaces are standard techniques [3, 19]; the case-specific payload is the simultaneous preservation of the named graded maps in (7).

From local seams to the produced values. The global seam proof is assembled from eight local families: the chosen old seam, the other inherited seams, the first and internal original seams, the last and internal mirror seams, and the positive and negative branch seams. An equivalence parametrizes the grafted diagram; its surjectivity reduces exhaustiveness to sum, endpoint, and Boolean cases. Consequently, the universal quantifier over new seams follows from a structural coverage theorem. In the inherited family, coordinate equivalences identify the new frames with the old named seam frame; grade-preservation lemmas make these equivalences suitable for the full predicate. The original solution then supplies the seam equation. The local proof therefore preserves the coordinate maps appearing in the seam equation.

²⁸See the constructor description in `Cutting/MarkedDiagram.lean`, lines 79–104.

²⁹See the definition of `SameFramedFlag`.

One local producer shows what this coordinate preservation requires. In `AtomicFrames`, ports enumerate finite sets of common labeled atoms. For a splitting vertex, two routes through the parent and child enumerations yield the same atom. Injectivity of the parent enumeration turns that atom identity into equality of the coordinate equivalences. The source transfers this equality to the comparison matrices, whose identical frames satisfy `SameFramedFlag` by reflexivity. Merging vertices reuse the splitting construction with reversed local data. Thus the compatibility starts from a common labeled object and reaches the named coordinate maps.³⁰ For an inherited seam, the later proof assembles six concrete inputs: grade preservation, the two frame identities, parent-word transport, child-word transport, and the original seam equation. These inputs expose the coordinate-level information still needed after rank preservation.³¹

There remains a second transport at the endpoint. The general `markedCut_allSeams` theorem proves the eight-family result for the assembled marked-cut data. The produced values use matrices rebased along the equality of the selected seam rank and its parent-facet rank. `rebaseUnit` performs this equality-induced change of matrix size. The endpoint proves that the chosen and rebased generator matrices agree, uses the mirror-rebasing theorem, and transports the resulting equality through the side-value constructor and then `valuesFromPorts`. Finally, `framedSeamEquations_of_values_eq` transfers the general seam predicate to the exact produced values. This last transport identifies the values in the theorem with the values returned by the constructor.

The boundary assumptions also distinguish two nearby statements. The all-seams endpoint has boundary indices b_0, i_0 , but requires no proof that this occurrence is the chosen seam. The later `producedMarkedSolution`, which assembles surface, seam, and vertex equations into a complete solution, additionally assumes

$$D.\text{boundarySide}\langle D.\text{bandCutParent}(q), b_0, i_0 \rangle = \langle q, \text{none} \rangle.$$

Here D is the original diagram, q is the selected seam, and `none` is Lean’s constructor for the absent optional index in a seam-side identifier. The extra hypothesis identifies the chosen boundary occurrence and is used for the surface component. The distinction identifies the selected theorem’s contribution to the larger solution construction. A common witness keeps the coordinates coherent, exhaustive local families cover every seam, and equality transport connects the local proofs to the final produced values.

4 A corpus comparison of proof-writing methods

The case studies explain how particular preservation arguments work. An exploratory survey asks how written proof methods are distributed across the wider collection. The baseline is `Mathlib` at the collection’s dependency revision. Its guidelines accommodate term and tactic proofs [45], making it a concrete comparison library with an established range of styles.

4.1 Sampling, extraction, and units of comparison

The sampling population contains 122,457 tracked modules strictly under `lean/OAI/`. From their paths we selected the 800 lowest SHA-256 ranks of a fixed seed followed by the path, before fetching their contents. The retained population list, seed, and Git blob identifiers reproduce membership. The baseline comprises all 8,070 mathematical modules under `Mathlib/` at the pinned revision, excluding infrastructure roots and umbrella files. The scanner extracts explicitly written, named

³⁰See the splitting, merging, and frame comparison proofs.

³¹See the inherited-seam proof.

`theorem` and `lemma` bodies: 13,309 in the OpenAI sample and 186,751 in Mathlib. Definitions, instances, anonymous examples, and declarations produced by macros are outside this unit.

Nested comments and strings are masked before locating declaration boundaries and the proof assignment. Extracted bodies cover 99.98% and 99.28% of detected named declarations, respectively; the remaining declarations, principally equation-style forms, are logged separately. Coverage here concerns detected source declarations. Validation uses forty stratified source inspections and eleven boundary checks, including `let` bindings in theorem types and inline scoped declarations. The validation record retains the inspected spans and two initial theorem-type binding errors, followed by their extraction fixes. This cohort tests known forms and boundary behavior; estimating error on unseen declarations would require a separate validation sample.

A feature is recorded once per body when a listed unqualified token occurs. Local claims include `have`, `suffices`, `obtain`, and `calc`; arithmetic and algebra include `omega`, `linarith`, `nlinarith`, `ring`, and related tactics. Restricted simplification records `simp`, `simpa`, or `simpall` followed by `only`. The retained script lists every token family. These indicators describe written proof features. Elaborated traces would additionally identify executed tactic nodes and their goal states.

Two summaries answer different questions. If module m has n_m extracted bodies and feature f occurs in c_{mf} of them, the pooled rate is $\sum_m c_{mf} / \sum_m n_m$. The module-balanced rate is $M^{-1} \sum_m c_{mf} / n_m$, over the M modules with at least one extracted body. Equivalently, select a proof-bearing module uniformly, then select one of its bodies uniformly. There are 784 and 7,348 such modules in the two corpora. We added module balancing after observing certificate concentration; both summaries are retained as exploratory analyses. The intervals in Table 2 resample the 784 OpenAI modules using 1,500 percentile-bootstrap replicates. The 95% intervals describe sampling variation conditional on extraction and feature definitions, with Mathlib treated as the retained census baseline.

Table 2. Module-balanced proof-body features (percent). The first three rows partition proof modes; subsequent features overlap. OpenAI intervals are exploratory 95% module-bootstrap intervals. Mathlib is the census baseline at the pinned revision.

Written feature	OpenAI	Mathlib	OpenAI interval
Outer tactic block	92.8	54.5	91.8–93.9
Term with a tactic subproof	0.5	5.7	0.2–0.9
Term without a tactic subproof	6.7	39.8	5.7–7.7
Rewrite or simplify	72.5	52.4	70.3–74.6
Local claims or calculation chains	61.1	21.0	58.8–63.6
Arithmetic or algebra tactics	38.2	2.7	35.5–40.8
Restricted simplification	38.5	13.0	36.4–40.9
Cases or induction	19.8	12.8	18.0–21.8
Finite decision	6.3	0.2	4.9–7.8

4.2 Separating subject composition, proof length, and method use

In the full comparison, the largest difference concerns outer proof form: the module-balanced share beginning in tactic mode is 92.8% in the OpenAI sample and 54.5% in Mathlib. The former also has more bodies containing local claims or calculation chains, arithmetic and algebra tactics, and restricted simplification. In inspected case proofs these features can work together: an intermediate claim records an estimate, rewriting exposes scalar expressions, and `nlinarith` or `omega` closes the resulting goal. The survey measures the prevalence of these devices; source reading explains particular instances of their use.

Subject composition matters. Restricting to 18 shared top-level domains retains 735 OpenAI proof-bearing modules, or 93.75% of that sample. Reweighting Mathlib’s within-domain module averages to those OpenAI domain shares gives tactic-mode rates of 93.0% versus 63.8%, local-claim rates of 61.4% versus 27.3%, arithmetic rates of 37.6% versus 7.1%, and restricted-simplification rates of 38.7% versus 15.8%. Extensionality and congruence become similar: 13.8% versus 13.1%. Directory labels provide a coarse subject control while retaining different theorem populations and declaration roles.

Figure 1 expresses these contrasts as differences from the Mathlib baseline in percentage points. The largest positive differences concern tactic-mode proofs, local claims, and arithmetic discharge. Term-only proofs are less frequent. Extensionality and congruence sit close to the baseline after domain standardization, showing that the difference is concentrated in particular proof devices.

Differences from the Mathlib baseline

Mean within-module shares of named theorem/lemma bodies

○ Full comparison ● Shared domains, standardized Mathlib

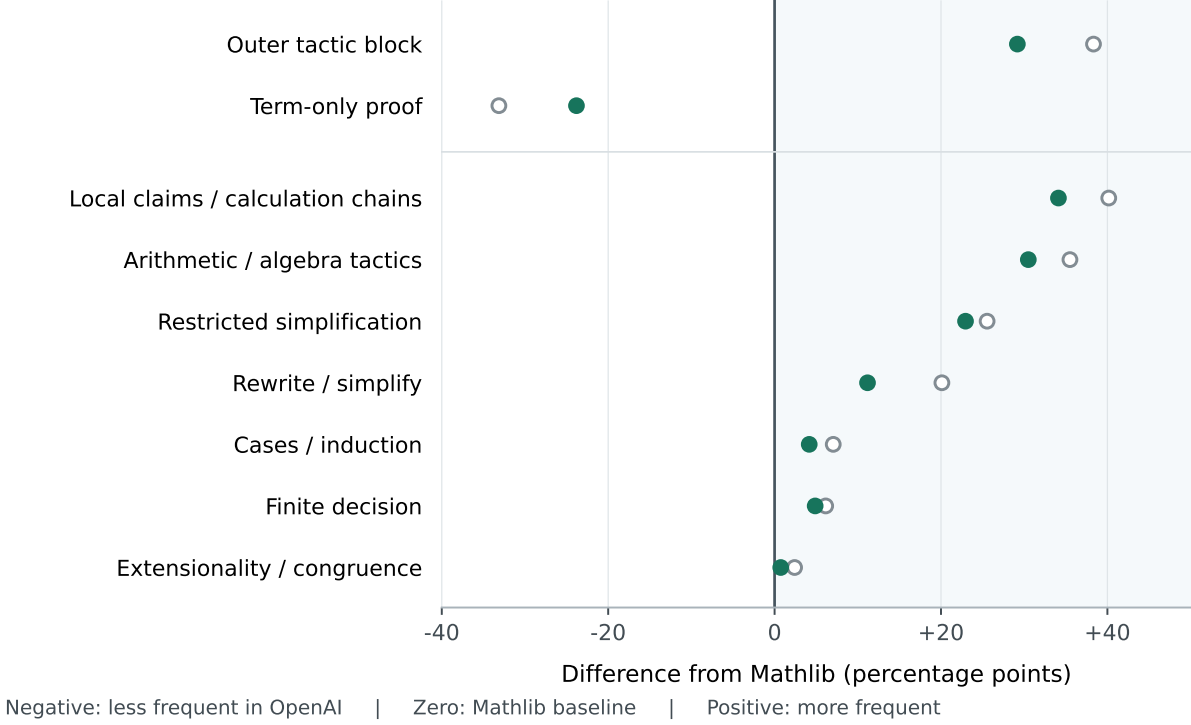


Figure 1. OpenAI minus Mathlib differences in written proof-feature prevalence, in percentage points. Zero denotes equal prevalence; positive values indicate greater prevalence in the OpenAI sample. Hollow markers compare all proof-bearing modules (784 OpenAI and 7,348 Mathlib). Green markers retain 735 OpenAI modules in 18 shared domains and reweight Mathlib to the OpenAI domain shares. Both markers lie at the same height for each feature and denote separate estimates of the difference from Mathlib. Both comparisons average within-module proof-body shares; features overlap except for the proof-mode categories. The figure shows point estimates. Table 2 reports absolute rates and 95% OpenAI sampling intervals for selected features in the full comparison. Table 3 adds joint subject and proof-length standardization.

Proof length changes the interpretation further. We jointly stratify by the shared directory domains and four bands of nonempty body lines: 1–3, 4–10, 11–30, and 31 or more. The underlying measure remains uniform selection of a proof-bearing module followed by a body within it. Within a cell, a module’s contribution is therefore proportional to the fraction of its bodies in that cell. Mathlib’s conditional feature rates are reweighted to the OpenAI cell probabilities. This preserves the module/body interpretation of Table 2. The 69 shared cells cover 93.75% of that OpenAI measure.

For a feature f , write $\mu_C(f \mid s)$ for its conditional probability in cell s under corpus C , and let w_s be the OpenAI cell probability normalized over shared cells. The adjusted difference in percentage points is

$$\Delta_f = 100 \sum_s w_s (\mu_{\text{OpenAI}}(f \mid s) - \mu_{\text{Mathlib}}(f \mid s)).$$

Table 3 compares this joint adjustment with the subject-only results. Its exploratory intervals use 1,500 resamples of OpenAI modules, recomputing their cell weights while keeping Mathlib’s cell

means fixed.

Table 3. OpenAI minus Mathlib differences in percentage points, with subject-only and joint subject/length standardization. Both use the module/body sampling measure. The 95% percentile-bootstrap intervals resample OpenAI modules with Mathlib cell means fixed. The joint comparison covers 69 cells and 93.75% of the OpenAI measure.

Written feature	Subject only	Subject + length	95% joint interval
Outer tactic block	+29.2	+10.4	9.3–11.5
Local claims or calculation chains	+34.1	+5.8	4.4–7.3
Arithmetic or algebra tactics	+30.5	+22.8	20.3–25.4
Restricted simplification	+22.9	+7.7	5.4–9.8
Finite decision	+4.9	+4.3	3.0–5.9

Arithmetic and algebra tactics retain a difference of 22.8 percentage points. The tactic-mode difference becomes 10.4 points and the local-claim difference 5.8, compared with 29.2 and 34.1 under subject-only adjustment. Thus proof-length composition is associated with much of the latter contrasts. Length may itself reflect how an argument is written, so conditioning on it changes the descriptive question. The two comparisons show both the overall profile within subjects and the remaining differences within broad subject/length cells.

Additional sensitivity checks remove each sampled directory-family group in turn. The OpenAI arithmetic rate remains between 37.0% and 38.7%, against 38.2% in the full module-balanced sample. These groups are rough proxies for developments; their exclusion shows that a single sampled group does not account for the arithmetic profile. The retained data also include the original pooled length-band comparison and alternative family weights. Matching analogous mathematical tasks and declaration roles would give a more specific baseline.

The listed general proof-search tokens, including `aesop`, `grind`, and `solve_by_elim`, have module-balanced rates of 0.07% in the OpenAI sample and 3.35% in Mathlib. Arithmetic tactics and simplification are themselves automation, and other search tactics or imported macros can escape this list. Similarly, `simp only` records explicit control of the rewrite set. Its maintenance value depends on the surrounding proof. Mathlib’s guidance treats squeezing terminal simplification calls as a tradeoff involving performance and fragility [45].

4.3 Why certificate batches change the apparent distribution

Concentration makes the distinction between modules and bodies essential. Ten sampled OpenAI modules contribute 48.2% of the extracted bodies; the largest alone contributes 14.9%. Consequently the pooled finite-decision rate is 48.9%, compared with the module-balanced 6.3%. Pooled rewrite/simplification rates are 41.5% versus Mathlib’s 48.8%, even though the module-balanced ordering in Table 2 is reversed. Pooled rates describe the produced body population; module averages describe practice under the module/body selection measure.

Reading these large files reveals an informative construction. The sampled planar-packing module `LowBlock36_10` contains 872 named bodies, 792 of which contain a finite decision.³² It defines exact integer interval data, checks individual entries with `decide +kernel`, and assembles

³²Source: `Analysis/PlanarPacking/Certificate/LowBlock36_10.lean`, lines 16–67.

them into finite sums. A sampled transonic barrier module chains Horner-step certificates with previously proved tail bounds.³³ The construction places exact data and small checks beneath soundness and assembly lemmas. Lean 4.34.1 documents `decide +kernel` as evaluating the decision instance by kernel reduction [42]. This mechanism explains how a large collection of short proofs serves as a numerical certificate for a later estimate.

The persistent arithmetic difference and the separate certificate layer provide the clearest empirical findings. The larger unadjusted contrasts also describe differences in proof form and length composition. Read alongside the cases, these results motivate a division-of-work account: preservation interfaces prepare mathematical obligations, and established Lean devices check the resulting local steps. Section 6.2 traces this interaction in particular proof bodies.

5 Worked examples: why the stronger interface matters

The first three examples separate the common-base pipeline into tape selection, output-law preservation, and an accuracy budget. The others isolate boundary locality, trace limits, named coordinates, and reuse of the actual ideal interface. Together they show why each preservation requirement is useful in the operation that follows it.

5.1 Injective tape positions preserve a uniform experiment

Proposition 1. *Let $p : \{0, \dots, m-1\} \rightarrow \{0, \dots, B-1\}$ be injective. If β is uniform on $\{0, 1\}^B$, then $(\beta(p(0)), \dots, \beta(p(m-1)))$ is uniform on $\{0, 1\}^m$. Every word has exactly 2^{B-m} preimages.*

Proof. Injectivity gives $m \leq B$. Prescribing the m selected bits leaves precisely $B - m$ coordinates free. Each assignment to these remaining coordinates determines one preimage. Thus every word has the same number of preimages, and its probability is $2^{B-m}/2^B = 2^{-m}$. $\square$

For the common-base machine, the relevant map is $p(i) = s + 3(m - 1 - i) + 1$. Reversal and multiplication by three preserve distinctness. For $i < m$, its value is less than B whenever $s + 3m + 1 \leq B$. This resource bound also accounts for the instructions surrounding the selected reads. The review’s Lean file proves these arithmetic facts for arbitrary natural s, m, B , including the empty-word case.

The index map is defined by

```
def physicalIndex (start m : Nat) (i : Fin m) : Nat :=
  start + 3 * (m - 1 - i.val) + 1
```

Listing 1. Lean definition of the physical tape index.

The proof uses the bounds stored in `Fin m` to justify subtraction arithmetic. This matters: natural-number subtraction is truncated, so injectivity would fail on unrestricted natural indices. The range type supplies exactly the bounds that make the index arithmetic valid.

For $m \geq 2$, if $p(0) = p(1)$, every word with different first two bits has no preimage. Marginal fairness of individual reads therefore leaves a separate joint-uniformity obligation. Proposition 1 explains why the source proves injectivity of the physical positions before transporting the success probability. The Python companion exhaustively checks 15 small choices of (s, m, B) against the exact fiber count.

³³Source: `MathematicalPhysics/Transonic/Certificates/Barrier59.lean`, lines 675–678.

5.2 All observables determine a finite law; one mean need not

Proposition 2. *Let μ, ν be probability measures on a finite set X . Then $\mathbb{E}_\mu \varphi = \mathbb{E}_\nu \varphi$ for every $\varphi : X \rightarrow \mathbb{R}$ if and only if $\mu = \nu$.*

Proof. For each $x \in X$, take $\varphi = \mathbf{1}_{\{x\}}$ to obtain $\mu(\{x\}) = \nu(\{x\})$. Summing singleton masses gives equality on every subset. The converse follows from the definition of finite expectation. $\square$

In a uniform finite experiment, these expectations are normalized sums with integer multiplicities. The Lean companion proves the corresponding count statement: equality of unnormalized sums against every natural-valued observable determines each multiplicity. If two sampling spaces have different cardinalities $N, N' > 0$, equality of their output laws instead means $N'c(x) = Nc'(x)$ for each output x . The companion also proves this cross-multiplied statement from equality of normalized observables. For example, multiplicities $(1, 1, 1)$ and $(2, 2, 2)$ describe the same law, although their unnormalized counts differ. It then checks the measures on $X = \{0, 1, 2\}$ with multiplicities

$$(1, 1, 1) \quad \text{and} \quad (0, 3, 0).$$

Both multiplicity vectors sum to three; after normalization, both distributions have mean one. Their probabilities of $\{0\}$ are $1/3$ and zero. Thus preservation of the expected numerical output would be insufficient for an accuracy event. The universal observable certificate in Section 3.1 supplies the needed event probability by substitution of the relative-error indicator.

This interpretation also explains why the source allows reindexing of draw outcomes. A permutation can change the result returned for a particular numbered bank without changing its distribution under uniform sampling. Law equality transports the probabilistic guarantee between these experiments. The compiler layer additionally preserves execution on each supplied bank.

5.3 From a bounded experiment to a finite success count

Tape uniformity and observable preservation compose with an implementation error bound. The resulting success count explains the accuracy statement in Section 3.1.

Proposition 3. *Let X be finite, adjoin a failure symbol $\perp$, and let U be an ideal X -valued output. Let V be a bounded implementation driven by a uniform m -bit word, taking values in $X \cup \{\perp\}$. Let $\eta, \delta_0 \geq 0$. For $\varphi : X \rightarrow [0, 1]$, set $\varphi(\perp) = 0$. Suppose*

$$\mathbb{E}\varphi(V) \geq \mathbb{E}\varphi(U) - \eta \quad \text{for every such } \varphi.$$

Let $A \subseteq X$ satisfy $\Pr(U \in A) \geq 1 - \delta_0$. If a machine obtains its word from fixed injective positions in a uniform B -bit tape and returns V on that word, then

$$\#\{\text{successful } B\text{-bit tapes}\} \geq (1 - \delta_0 - \eta)2^B,$$

where success means returning an element of A .

Proof. Choose $\varphi = \mathbf{1}_A$ to obtain $\Pr(V \in A) \geq 1 - \delta_0 - \eta$. Proposition 1 shows that the tape projection has the required uniform word law. Multiplying its success probability by 2^B gives the count. $\square$

For a single accuracy event, only its indicator is needed; the all-observables hypothesis makes the estimate reusable for other events. In the source's bounded draw-tree implementation, the fit condition requires at most M successive draws and $H \geq \lceil \log_2 a \rceil$ trial bits for every draw arity a .

With D trials per draw, $\eta = M2^{-D}$, with failed runs assigned observable value zero.³⁴ For a desired failure budget $0 < \delta < 1$, choosing $\delta_0 \leq \delta/2$ and $M2^{-D} \leq \delta/2$ suffices. When $M > 0$, one may take $D \geq \max\{0, \lceil \log_2(2M/\delta) \rceil\}$. The tape-length and execution proofs supply the remaining inputs. The final guarantee therefore combines a probabilistic loss budget, a uniform gathered word, and deterministic execution on that word.

A tight finite example joins the layers. Let the ideal probabilities of a, b be $6/8, 2/8$, and assign the eight implementation words to five copies of a , two of b , and one failure. The observable loss is exactly $\varphi(a)/8$, hence at most $\eta = 1/8$. For $A = \{a\}$, $\delta_0 = 1/4$ and the lower bound is $5/8$. Embedding the three-bit word at positions 7, 4, 1 in a ten-bit tape gives exactly $5 \cdot 2^7 = 640$ successful tapes out of 1024. The Python companion enumerates those tapes and all event indicators. This finite fixture makes every layer of the composed bound directly countable.

5.4 Boundary values and ambient derivatives

The Mahler case uses local equality to identify ambient derivatives. On $\mathbb{R}^2$, the one-forms $\omega = 0$ and $\eta = (x^2 + y^2 - 1) dx$ agree at every point of the unit circle. Their ambient exterior derivatives differ: $d\eta = -2y dx \wedge dy$, which is nonzero at $(0, 1)$. Equality on a neighborhood identifies the ambient derivatives. The Mahler cutoff argument uses this sufficient condition to transfer its ambient derivative; agreement on tangent arguments suffices for the later comparison of integration densities. Intrinsic differentiation after pullback to the circle uses different data: both pulled-back forms, and their exterior derivatives, vanish. Mathlib's differential-form interface explicitly uses the derivative of a form-valued field [44].

5.5 A cut-sensitive limit that is not pointwise

Consider the commutative C^* -algebra c_0 of complex sequences tending to zero, with pointwise operations and the supremum norm. On its positive cone define

$$\tau(a) = \sum_{j=1}^{\infty} a_j \in [0, \infty].$$

The trace is additive, positively homogeneous, and tracial. It is lower semicontinuous because it is the supremum of its continuous finite partial sums. Finite-support sequences belong to $\mathcal{N}_\tau$, so the closed norm-closed ideal I_τ in (2) is all of c_0 .

For every positive $a \in c_0$ and $\varepsilon > 0$, the sequence $(a - \varepsilon)_+$ has finite support. Consequently

$$\lim_{r \downarrow 0} r\tau((a - \varepsilon)_+) = 0, \quad 0 \leq \liminf_{r \downarrow 0} r\tau(a).$$

These are precisely the two inequalities in (5) with limit trace $\iota_{c_0} = 0$. But for $a_j = 1/j$ one has $\tau(a) = \infty$, hence $r\tau(a) = \infty$ for every $r > 0$. The pointwise limit at this element is infinity. The cut-regularized limit still recovers this norm-closed ideal.

The example instantiates the cut-regularized convergence studied by Elliott et al. [18]. Closure admits elements with infinite trace, while every fixed positive cut is controlled by a finite value. Replacing the source's convergence predicate by pointwise convergence would therefore alter the theorem.

³⁴See `CommonBases.lean, run_lower`.

5.6 Equal flags and different named coordinates

The character-variety case preserves more than the flag of subspaces. Choose a basis vector v of a one-dimensional vector space over a field of characteristic zero, with its coordinate in grade zero. The frames $e(u) = uv$ and $f(u) = 2uv$ induce the same flag $0 \subset \text{span}(v)$, while $e^{-1}(v) = 1$ and $f^{-1}(v) = 1/2$. The all-seams predicate must therefore retain the named graded maps when those maps occur in the seam equation. The coordinate equation depends on the specified maps on the graded piece. This is the elementary obstruction that the surface construction addresses with named frame data.

5.7 Composing the actual ideal correspondence

The trace case permits a direct reuse of its formal interface. Write $T(A)$ for the extended-trace cone, and let $e : T(A) \simeq T(B)$ and $f : T(B) \simeq T(C)$ be additive equivalences. Let Φ_e be the source's induced order isomorphism of closed ideals. Its weight identity gives

$$\iota_{\Phi_e(I)} = e(\iota_I).$$

Applying this identity successively yields

$$\iota_{\Phi_f(\Phi_e(I))} = f(e(\iota_I)) = \iota_{\Phi_{f \circ e}(I)}.$$

Injectivity of ideal weights therefore proves $\Phi_{f \circ e}(I) = \Phi_f(\Phi_e(I))$. This compositional law uses the additive equivalences alone; the convergence and scalar fields become relevant when identifying the ideals of particular traces.

The accompanying `UpstreamReuse.lean` imports `OAI.Analysis.TraceCone.IdealTransport`. Its proof invokes the upstream ideal-weight identity three times and then uses definitional equality for composition. The corollary shows the practical value of the identifying equation: composition of the reconstructed ideal maps follows directly from the upstream API. The ideal reconstruction is reused through its specification.

6 Discussion: interfaces and the division of proof work

6.1 What the mathematical comparison reveals

The downstream operation determines what a preservation statement must expose. Transferring an accuracy guarantee requires the relevant event probability; equality of output laws supports all such transfers. A named-coordinate equation uses the coordinate map. Ambient differentiation uses local information about a field. A sharp norm estimate uses the normalization and weights of the preceding bounds. Exact data, laws, estimates, and local differential information can therefore appear together in one development.

The most informative transitions occur when the next operation changes these requirements. The common-base proof preserves a distribution through reindexing, then execution on each supplied bank. Mahler identifies form-valued fields for differentiation, then uses tangential agreement for boundary integration. Trace-cone reconstruction obtains ideal order from addition, then uses convergence and scaling to identify the ideals of a particular trace. Following the next operation explains why each interface has its particular fields.

Implementation details sharpen the comparison. Encoded width and numerical attributes answer different complexity questions in the compiler. The graph restriction identity restores

the counting norm when local assignment estimates are summed. Image filters extend a carrier-indexed convergence property to arbitrary nets. These choices expose mathematical data through lemmas that later constructions can reuse. The compiled ideal-composition example in Section 5.7 demonstrates this reuse directly.

Two cases show how a local calculation controls a global estimate. The graph’s total-two table isolates an exceptional direction whose aggregate energy is paid for by switch gains. In Crouzeix, rectangular compressions and two orders of polynomial multiplication produce compatible energy bounds. The final scalar cancellation is short because the earlier constructions have supplied its exact constants and signs.

These patterns have substantial antecedents. Verified compilers establish semantic preservation between source and implementation [29, 30]; probabilistic program logics reason about distributions and observations [6, 7]. Algebraic hierarchy design and specification-driven collaboration make interfaces part of mathematical construction [3, 13, 19]. The present comparison identifies what those interfaces preserve in the selected proofs and how their composition supplies a later calculation’s inputs.

6.2 Tactic use and the division of proof work

The corpus survey adds a measured comparison of written proof practice. After joint subject and proof-length standardization, arithmetic and algebra tactics remain more prevalent by 22.8 percentage points. Tactic mode and local claims retain differences of 10.4 and 5.8 points, respectively. The larger subject-only contrasts in Figure 1 also reflect proof-length composition; Table 3 makes that distinction explicit. Arithmetic emphasis is the clearest persistent difference among these features.

The common-base fold-cost proof shows how such local automation fits into a semantic argument. Execution identities justify applying the induction hypothesis to the remaining fold. The safety invariant and width bound supply its premises. The following consecutive source lines then apply that hypothesis and close the cost inequality:³⁵

```
have hh := ih z' y' e (hs.2 z' hrun) hw' ht
simp only [List.length_cons, Nat.add_mul, Nat.one_mul]
omega
```

Listing 2. Lean source excerpt: fold induction followed by arithmetic discharge.

Here `hs.2` supplies safety after the accumulator update, and `hw'` bounds the remaining list’s width. Expanding the successor length exposes shared product terms. Although the displayed cost bound contains multiplication, the remaining comparison fits linear natural-number arithmetic. The semantic and size arguments prepare the obligation that `omega` checks.

³⁵Source: `CommonBases.lean`, lines 16880–16907; excerpt, lines 16905–16907.

The Crouzeix proof supplies a parallel example in analysis. Fourier and positive-block estimates have already produced bounds with the same energy a and norm γ . Their closing scalar argument is:³⁶

```

have hb : 2*a ≤ 8*a/γ^2 := by
  simp only [mul_div_assoc] at hPE hQE ⊢
  linarith
have hmul := (le_div_iff0 (sq_pos_of_pos hγ)).mp hb
change γ ≤ 2
have hg2 : γ^2 ≤ 4 := by nlinarith [hapos]
nlinarith

```

Listing 3. Lean source excerpt: the closing Crouzeix energy calculation.

The first tactic combines the available energy inequalities. Positivity of γ then permits denominator clearing, and positivity of a permits the final cancellation. The choice of compressions, Fourier weights, and zero-mode identities creates the relations that these arithmetic steps certify.

Exact numerical certificates exhibit a further division of work. Their finite checks verify retained interval data; assembly and soundness lemmas explain why those checks imply the required estimate. Across these examples, representation design and local automation meet at an explicit obligation with proved hypotheses. The survey establishes a broader method profile, while the case tracing explains mechanisms within completed source proofs. Testing effects on completion rates, development effort, or maintenance would require matched tasks and corresponding outcome measurements.

6.3 Proof generation, simplification, and mathematical explanation

The comparison identifies differences between completed source corpora. Mathlib is a collaborative library with varied authoring histories, rather than a controlled sample of unaided human proofs. These data therefore cannot establish a distinct AI reasoning process, an advantage in completion time, or a superior method of doing mathematics. They suggest a workflow hypothesis: automation may be especially effective when proved interfaces expose many local obligations in forms handled by existing tactics. The relevant conditions—available representations, library coverage, and checkable local goals—can vary within a subject as well as between subjects. Interface construction remains part of the work whose cost must be measured.

Obtaining a checked proof and obtaining an economical explanation are separate objectives. ProofOptimizer reports substantial token reductions in generated Lean proofs, using Lean to check replacements against the original theorem statements [25]. Lean Refactor studies proof length, compilation cost, and compatibility across library versions as distinct optimization objectives [31]. These results indicate scope for refactoring on the evaluated benchmarks; they do not measure redundancy in the six developments examined here. Shorter tactic scripts can also invoke heavier automation. Source length, proof-checking cost, dependency structure, and explanatory clarity should consequently be assessed separately.

Together these observations motivate a research direction: connect proof generation to compression, reusable interfaces, and mathematical explanation. A useful interface may amortize its construction cost over many later results, while a concise final proof may conceal substantial setup. Claims of improved mathematical practice require evidence about total effort, generality, downstream reuse, and reader understanding, alongside correctness. The present findings locate possible mechanisms and evaluation targets without establishing those improvements.

³⁶Source: `DirectCrouzeix/DomainCore.lean`, lines 105–117; excerpt, lines 111–117.

6.4 Using the comparison in a new formalization

Begin with the quantity the next theorem consumes: an event probability, a coordinate equation, a derivative, or an inequality. Identify what a proposed representation changes, such as multiplicities, coordinate order, multiplication order, or the neighborhood of definition. The preservation statement should connect those changed data to that specific quantity.

The worked examples make the choice concrete. An event indicator is sufficient to transfer one accuracy probability, while a full output-law identity supports reuse for other events. Neighborhood agreement identifies ambient derivatives; tangential agreement can suffice for the later integral. A named seam equation needs its coordinate maps. Proposition 3 combines several interfaces in one count: implementation loss, tape uniformity, and deterministic execution jointly yield the success guarantee.

Next, follow how the hypotheses are supplied for the actual inputs. Mahler’s strip maps construct the analytic data, and Crouzeix’s outer domains provide strict containment and the required estimates. In the marked-surface construction, one splitting witness supplies both the diagram and its coordinate identifications. The reusable design consists of a precise local model, a preservation relation, and a construction of the data required to apply it.

Finally, inspect the local goals exposed by that design. A width invariant can reduce a compiler cost step to natural-number arithmetic. Positive denominators and nonzero factors can prepare an analytic cancellation. Exact interval data can make a numerical obligation decidable. The interface proofs assign established automation a concrete part of the larger mathematical argument to check.

6.5 Scope and further work

The six cases were selected for interfaces that could be traced in depth. Their mathematical comparisons are complemented by the separate uniform module sample. Joint domain/length controls and directory-family sensitivity checks sharpen the survey’s descriptive interpretation. A comparison of analogous declaration roles and mathematical tasks would further separate library composition from choices about proof organization. Elaborated tactic traces and a held-out extraction-validation cohort would connect the lexical measurements more directly to executed proof steps. A prospective comparison could cross AI assistance with availability of proved representation interfaces on matched tasks, retaining unsuccessful attempts. Relevant outcomes include time to a checked proof, human intervention, inference and checking costs, later reuse, and blinded expert assessments of clarity. This would test the workflow hypothesis without treating completed-proof length as a causal control.

Source reading identifies the inspected declarations, their hypotheses, and the constructions connecting them. Compilation of the worked examples and the three-module trace-cone reuse experiment supplies additional checking evidence at those targets. Rebuilding the remaining case endpoints and examining their mathematical arguments would extend this assessment. Endpoint statements and support results retain their distinct roles throughout the comparison.

For readers of substantial Lean developments, the practical route is to locate the next operation, identify what it consumes, and follow the theorem preserving that information. For formalizers, the cases show how interface design prepares local proof obligations. The corpus comparison adds evidence about which written devices are prevalent in this collection, with arithmetic emphasis persisting under coarse joint controls. The review brings these two levels together: the mathematical content of preservation statements and the local proof work they support.

Data and code availability

The accompanying reproduction repository (release 1.6.0) provides the manuscript source, data, and analysis code [21]. The source package contains the LaTeX manuscript, reference records, declaration ledgers, original Lean examples, and Python checks. Five elementary Lean lemmas import only `Lean`. The upstream reuse experiment imports `OAI.Analysis.TraceCone.IdealTransport` and compiles its three-module dependency chain from unchanged source bytes. Both use Lean 4.34.1; the latter uses Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Compiler logs retain the reported axioms. Exact Python checks enumerate tape fibers and Boolean matrix counts. The README gives reproduction commands and identifies the external dependencies.

The method-survey directory retains the complete path population, all 800 sampled source modules and their hashes, extraction tables, feature definitions, bootstrap summaries, and forty inspected validation cases. The joint comparison additionally retains cell weights, conditional rates, intervals, and the generated manuscript table. Baseline profiling requires the pinned Mathlib source checkout; the lexical survey reads source without elaborating the developments. Public manuscript source links resolve to the reviewed repository revision.

The source inventory is the complete Git tree of `openai/math`, inspected on 8 October 2026.³⁷ It contains 122,875 files with the `.lean` extension, including supporting modules. The package retains 546 source artifacts, including the `OAI` import closures for the six selected cases; twenty-one detailed records document the additional proof-body analysis. The separate formalization catalogue lists 173 source records and 200 main-result bindings [34]. These inventories count different objects and should be read alongside their stated units.

The collection also supplies Comparator challenge configurations [36]. The inspected checker implements statement comparison, dependency and axiom inspection, kernel replay, and an optional external checker [10].³⁸ The distinction between proof construction and checking has a long history [5, 15]; Lean typechecker verification supplies a contemporary example [11]. Library maintenance addresses consistency across a growing development [46]. Interpreting the mathematics additionally requires reading the formal objects and hypotheses, the task pursued by the case studies here.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

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³⁷Reviewed revision: `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Public source links refer to that revision.

³⁸Checker source revision: `ca04cfc72b550331658ec314bf47685281bfd4bf`. This records the infrastructure source inspected; the collection’s installation instructions do not specify that revision.

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